

An Internship Report

Of

Python Programming virtual internship

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)



By

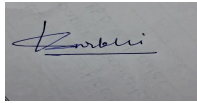
Roll Number: **25SCS1003002242**

Name : **LAXMI SIDDHARTH SURBHI**

Batch : **2026-27**

CANDIDATE'S DECLARATION

I, Laxmi Siddharth Surbhi do hereby declare that the internship report titled "Python Programming" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.



Signature

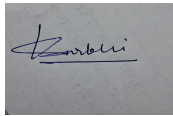
Student Name - Laxmi Siddharth Surbhi

Roll No. - 25SCS1003002242

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

A small, square, grayscale image of a handwritten signature. The signature is written in a cursive, flowing style, appearing to read 'Laxmi' followed by a surname that is partially obscured by a horizontal line.

Signature

Student Name - Laxmi Siddharth Surbhi

Roll No. - 25SCS1003002242

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3. INTERNSHIP COMPLETION CERTIFICATE

Internship Offer Letter



CodSoft

INTERNSHIP OFFER LETTER

Date : 26/07/2026

ID: BY26RY223849

Dear **Laxmi Siddharth Surbhi**,

We would like to congratulate you on being selected for the **“Python Programming”** virtual internship position with **“CodSoft”**. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **01 August 2026 to 31 August 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Sincerely

Ruman Arshad
Program Manager



ISO Number - TSNUK79053



MSME Regd



contact@codsoft.in

www.codsoft.in

4. PROJECT DESCRIPTION

4.1 Introduction

Python is a high-level programming language widely used for developing applications because of its simple syntax and wide range of libraries and modules. It supports rapid development and is suitable for creating both basic and advanced applications.

The Python Programming Internship involved the development of three practical applications: To-Do List Application, Password Generator, and Calculator Application. Each project was developed to apply fundamental Python concepts in a practical environment.

The projects provided hands-on experience with programming logic, functions, user input, operators, modules, graphical user interfaces, testing, and debugging. Tkinter was used to create interactive graphical interfaces for the applications.

The overall project work demonstrates how basic programming concepts can be combined to develop useful and functional applications.

4.2 Organization Profile

CODSOFT is an organization that provides internship opportunities focused on practical learning and skill development. The internship programs provide students with project-based tasks that allow programming concepts to be applied to practical applications.

The Python Programming Internship focused on developing applications using Python and understanding the basic process of application development. The assigned projects provided practical exposure to coding, interface development, testing, debugging, and project organization.

The major projects completed during the internship were:

- To-Do List
- Calculator
- Password Generator

These projects provided an opportunity to strengthen Python programming skills through practical implementation.

4.3 Problem Statement

The projects were developed to address simple problems that can be solved through programming.

To-Do List

Managing multiple daily tasks manually can make it difficult to keep track of pending and completed activities. A simple digital application can make task organization easier by providing options to add, view, complete, and remove tasks.

Calculator

Basic mathematical operations are frequently required in academic and everyday activities. An interactive calculator can provide a quick and convenient way to perform common mathematical calculations.

Password Generator

Creating passwords manually may result in repetitive or predictable combinations. A password generation tool can simplify the process by producing random passwords using different characters according to the required length.

4.4 Project Objectives

The major objectives of the internship projects were:

- To develop practical applications using Python.
- To strengthen understanding of Python programming fundamentals.
- To improve logical thinking and problem-solving skills.
- To understand the use of functions, operators, lists, and modules.
- To develop interactive graphical interfaces using Tkinter.
- To implement user input and event handling.

Project-Specific Objectives

To-Do List:

To provide a simple application for creating and managing daily tasks.

Calculator:

To perform common arithmetic operations through an interactive interface.

Password Generator:

To generate random passwords based on user-defined requirements.

4.5 Scope of the Project

The developed applications provide basic functionality and can be further improved with additional features.

To-Do List

The application can be extended by adding task priorities, due dates, reminders, categories, search options, and database storage.

Calculator

The calculator can be expanded with scientific calculations, calculation history, memory functions, keyboard support, and advanced mathematical operations.

The current projects provide a foundation for developing more advanced applications in the future.

Password Generator

Future improvements may include a password strength indicator, copy-to-clipboard functionality, password history, and additional customization options.

4.6 Technologies and Tools Used

Python:

Python was used as the primary programming language for developing all three applications. It

was used to implement the application logic, functions, calculations, user input, and required processing.

Tkinter:

Tkinter was used to design the graphical user interfaces of the applications. It provided interactive components such as windows, buttons, labels, and input fields.

Visual Studio Code:

Visual Studio Code was used as the development environment for writing, executing, and debugging the Python programs. It provided a convenient environment for managing the project source files.

GitHub:

GitHub was used to store and organize the source code of the internship projects. It also provided a platform for maintaining the project repository and sharing the completed work.

4.7 System Architecture

The developed applications follow a **three-layer architecture** consisting of the **Presentation Layer, Application Layer, and Output Layer**. The architecture separates user interaction from program processing and result generation, making the applications easier to understand and manage.

1. Presentation Layer

This layer represents the graphical interface through which the user interacts with the application. It contains elements such as buttons, labels, input fields, and display areas created using Tkinter.

2. Application Layer

This layer contains the main Python programming logic. It receives the user input and performs the required operations using functions, conditions, operators, lists, and Python modules.

3. Output Layer

This layer displays the processed information to the user through the graphical interface. The output varies according to the application, such as an updated task list, generated password, or calculated result.

4.8 Methodology

Stage	Description
Requirement Analysis	The requirements and expected functionality of each project were studied before development.
Planning	The main features, application structure, and required interface components were planned.
Implementation	Python code was written to implement the required functionality of each application.
GUI Development	Tkinter was used to create the graphical interface and interactive components.
Testing	Different inputs and operations were tested to verify that the applications worked correctly.
Debugging	Errors identified during testing were analyzed and corrected through code modifications.
Documentation	The completed projects and source codes were organized and maintained in the GitHub repository.

4.9 Expected Outcomes

The projects are expected to provide practical understanding of Python programming and application development.

The major expected outcomes include:

- Better understanding of Python programming fundamentals.

- Practical experience in developing GUI applications.
- Improved logical and problem-solving abilities.
- Understanding of functions, operators, lists, strings, and modules.
- Experience in handling user input and application events.
- Improved debugging and testing skills.
- Basic experience with GitHub and project organization.

4.10 Certificates of Completion and Communication Mail Proof

This section presents the supporting documents related to the successful completion of the Python Programming Internship. The attached documents provide evidence of internship participation, project completion, and submission of the assigned tasks.

The **Internship Completion Certificate** issued by CODSOFT is included as official proof of completion of the internship. Relevant **communication emails** related to the internship and assigned tasks are also attached for verification.

CODSOFT AUGUST TASK SUBMISSION FORM(BATCH C15)

Thanks for submitting your contact info!
Certificate will be release on 01 SEPTEMBER
2026 at approximately 10:00 PM Night

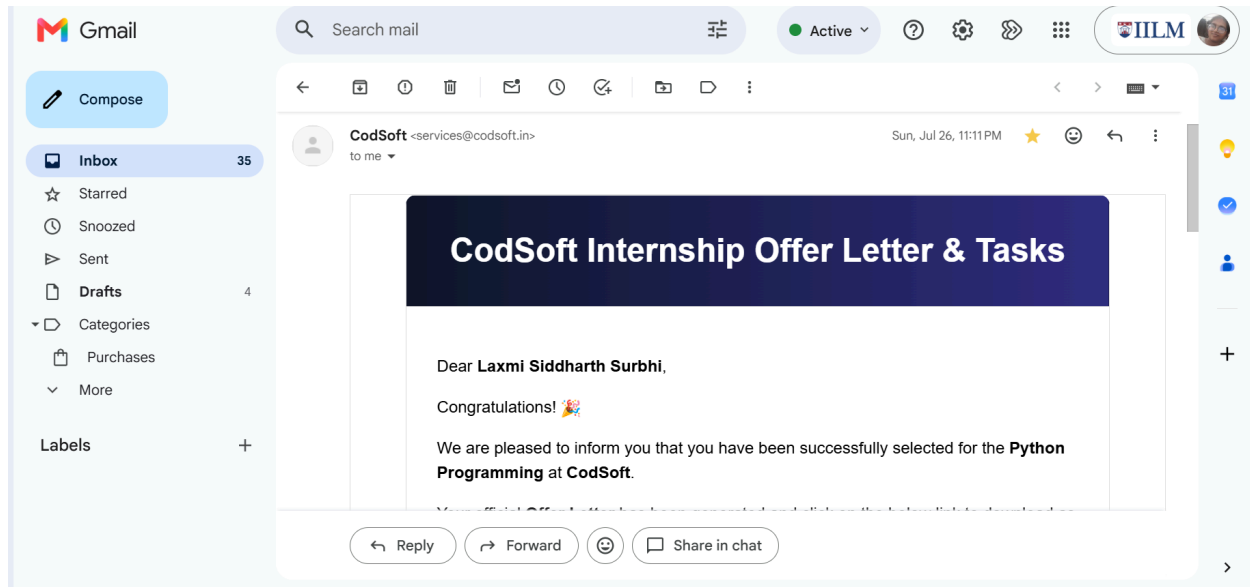
Certificate will be release on 01 SEPTEMBER
2026 at approximately 10:00 PM Night

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